

## Fabian Gittins — Publications

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| CONTACT<br>INFORMATION        | Institute for Gravitational and Subatomic Physics<br>Princetonplein 1, Utrecht University<br>3584 CC Utrecht, The Netherlands                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <a href="mailto:f.w.r.gittins@uu.nl">f.w.r.gittins@uu.nl</a><br><a href="https://fgittins.github.io">fgittins.github.io</a> |
| PUBLICATION<br>SUMMARY        | Full list of publications can be found on <a href="#">Google Scholar</a> , <a href="#">INSPIRE-HEP</a> and <a href="#">NASA ADS</a> .<br><br><b>h-index</b> —16 (according to Google Scholar) or 14 (according to INSPIRE-HEP). Both include collaboration papers.<br><br><b>Top five cited</b> —Excluding long-author-list papers. Citation counts according to Google Scholar. <ol style="list-style-type: none"><li>1. <b>Gittins, F.</b>, Andersson, N., Jones, D. I., <i>Modelling neutron star mountains</i>, <a href="#">Mon. Not. R. Astron. Soc. <b>500</b>, 5570 (2021)</a> [<a href="#">arXiv:2009.12794</a>]. (82 citations)</li><li>2. <b>Gittins, F.</b>, Andersson, N., <i>Modelling neutron star mountains in relativity</i>, <a href="#">Mon. Not. R. Astron. Soc. <b>507</b>, 116 (2021)</a> [<a href="#">arXiv:2105.06493</a>]. (75 citations)</li><li>3. <b>Gittins, F.</b>, Andersson, N., Pereira, J. P., <i>Tidal deformations of neutron stars with elastic crusts</i>, <a href="#">Phys. Rev. D <b>101</b>, 103025 (2020)</a> [<a href="#">arXiv:2003.05449</a>]. (52 citations)</li><li>4. Pereira, J. P., Bejger, M., Andersson, N., <b>Gittins, F.</b>, <i>Tidal deformations of hybrid stars with sharp phase transitions and elastic crusts</i>, <a href="#">Astrophys. J. <b>895</b>, 28 (2020)</a> [<a href="#">arXiv:2003.10781</a>]. (44 citations)</li><li>5. <b>Gittins, F.</b>, <i>Gravitational waves from neutron-star mountains</i>, <a href="#">Class. Quantum Gravity <b>41</b>, 043001 (2024)</a> [<a href="#">arXiv:2401.01670</a>]. (37 citations)</li></ol> |                                                                                                                             |
| COLLABORATION<br>PUBLICATIONS | Since 2026, I have been a co-author on refereed LIGO-Virgo-KAGRA and Einstein Telescope Collaboration publications. I list only non-collaboration articles below.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                             |
| SUBMITTED<br>PUBLICATIONS     | <p>[28] Yu, H. et al., <i>Nonlinear hydrodynamics in spinning neutron stars: Theoretical universal relations and equilibrium solutions</i> [<a href="#">arXiv:2607.07943</a>].</p> <p>[27] Negri, L. et al., <i>Impact of the Einstein Telescope’s duty cycle on the estimation of binary black holes parameters</i> [<a href="#">arXiv:2606.19201</a>].</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                             |
| ACCEPTED<br>PUBLICATIONS      | <p>[26] <b>Gittins, F.</b> et al., <i>Detecting Tidal Resonances in Binary Neutron Stars</i>, <a href="#">Phys. Rev. Lett. <b>137</b>, 011101 (2026)</a> [<a href="#">arXiv:2606.06376</a>].</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                             |
| REFEREED<br>PUBLICATIONS      | <b>Research articles</b> <p>[25] Andersson, N., Counsell, A. R., <b>Gittins, F.</b>, Ghosh, S., <i>Tidal response of a relativistic star</i>, <a href="#">Phys. Rev. D <b>113</b>, 064051 (2026)</a> [<a href="#">arXiv:2511.05139</a>].</p> <p>[24] Yin, S., Andersson, N., <b>Gittins, F.</b>, <i>A post-Newtonian approach to neutron star oscillations</i>, <a href="#">Class. Quantum Gravity <b>42</b>, 235002 (2025)</a> [<a href="#">arXiv:2504.06918</a>].</p> <p>[23] Pnigouras, P., Andersson, N., <b>Gittins, F.</b>, Counsell, A. R., <i>Dynamical neutron star tides: the signature of a mode resonance</i>, <a href="#">Mon. Not. R. Astron. Soc. <b>542</b>, 1375 (2025)</a> [<a href="#">arXiv:2508.06416</a>].</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                             |

- [22] Counsell, A. R., **Gittins, F.** et al., *Interface modes in inspiralling neutron stars: A gravitational-wave probe of first-order phase transitions*, **Phys. Rev. Lett.** **135**, 081402 (2025) [arXiv:2504.06181].
- [21] **Gittins, F.**, Andersson, N., Yin, S., *Perturbation theory for post-Newtonian neutron stars*, **Class. Quantum Gravity** **42**, 135014 (2025) [arXiv:2503.03345].
- [20] **Gittins, F.**, Andersson, N., *Neutron-star seismology with realistic, finite-temperature nuclear matter*, **Phys. Rev. D** **111**, 083024 (2025) [arXiv:2406.05177].
- [19] **Gittins, F.** et al., *Problematic systematics in neutron-star merger simulations*, **Phys. Rev. D** **111**, 023049 (2025) [arXiv:2409.13468].
- [18] Counsell, A. R., **Gittins, F.** et al., *Neutron star g modes in the relativistic Cowling approximation*, **Mon. Not. R. Astron. Soc.** **536**, 1967 (2025) [arXiv:2409.20178].
- [17] Counsell, A. R., **Gittins, F.**, Andersson, N., *The impact of nuclear reactions on the neutron-star g-mode spectrum*, **Mon. Not. R. Astron. Soc.** **531**, 1721 (2024) [arXiv:2310.13586].
- [16] Pnigouras, P., **Gittins, F.** et al., *The dynamical tides of spinning Newtonian stars*, **Mon. Not. R. Astron. Soc.** **527**, 8409 (2024) [arXiv:2205.07577].
- [15] Beri, A. et al., *AstroSat and NuSTAR observations of XTE J1739-285 during the 2019-2020 outburst*, **Mon. Not. R. Astron. Soc.** **521**, 5904 (2023) [arXiv:2303.13085].
- [14] **Gittins, F.** et al., *Modelling Neutron-Star Ocean Dynamics*, **Universe** **9**, 226 (2023) [arXiv:2304.05413].
- [13] **Gittins, F.**, Andersson, N., *The r-modes of slowly rotating, stratified neutron stars*, **Mon. Not. R. Astron. Soc.** **521**, 3043 (2023) [arXiv:2212.04892].
- [12] Andersson, N., **Gittins, F.**, *Formulating the r-mode Problem for Slowly Rotating Neutron Stars*, **Astrophys. J.** **945**, 139 (2023) [arXiv:2212.04837].
- [11] Andersson, N., **Gittins, F.** et al., *Building post-Newtonian neutron stars*, **Class. Quantum Gravity** **40**, 025016 (2023) [arXiv:2209.05871].
- [10] Riley, J. et al., *Rapid Stellar and Binary Population Synthesis with COMPAS*, **Astrophys. J. Suppl. Ser.** **258**, 34 (2022) [arXiv:2109.10352].
- [9] **Gittins, F.**, Andersson, N., *Modelling neutron star mountains in relativity*, **Mon. Not. R. Astron. Soc.** **507**, 116 (2021) [arXiv:2105.06493].
- [8] **Gittins, F.**, Andersson, N., Jones, D. I., *Modelling neutron star mountains*, **Mon. Not. R. Astron. Soc.** **500**, 5570 (2021) [arXiv:2009.12794].
- [7] **Gittins, F.**, Andersson, N., Pereira, J. P., *Tidal deformations of neutron stars with elastic crusts*, **Phys. Rev. D** **101**, 103025 (2020) [arXiv:2003.05449].
- [6] Pereira, J. P., Bejger, M., Andersson, N., **Gittins, F.**, *Tidal deformations of hybrid stars with sharp phase transitions and elastic crusts*, **Astrophys. J.** **895**, 28 (2020) [arXiv:2003.10781].
- [5] **Gittins, F.**, Andersson, N., *Population synthesis of accreting neutron stars emitting gravitational waves*, **Mon. Not. R. Astron. Soc.** **488**, 99 (2019) [arXiv:1811.00550].

#### Review articles

- [4] **Gittins, F.**, *Gravitational waves from neutron-star mountains*, **Class. Quantum Gravity** **41**, 043001 (2024) [arXiv:2401.01670].

#### Software articles

- [3] Riley, J. et al., *COMPAS: A rapid binary population synthesis suite*, *J. Open Source Softw.* **7**, 3838 (2022).

#### Conference proceedings

- [2] Thomas, A. Stevenson, E., **Gittins, F.** et al., *Galactic Archaeology with TESS: Prospects for Testing the Star Formation History in the Solar Neighbourhood*, *EPJ Web Conf.* **160**, 05006 (2017) [[arXiv:1610.08862](#)].

#### WHITE PAPERS

- [1] Dietrich, T. et al., *ESO Expanding Horizon White Paper: Revealing the properties of matter at supranuclear densities with gravitational waves* [[arXiv:2512.16971](#)].